www.tntextbooks.in

9.12.1 Introduction to fundamental Data types:

Fundamental (atomic) data types are predefined data types available with C++. There are five fundamental data types in C++: **char, int, float, double** and **void**. Actually, these are the keywords for defining the data types.

(1) int data type:

Integers are whole numbers without any fraction. Integers can be positive or negative. Integer data type accepts and returns only integer numbers. If a variable is declared as an **int**, C++ compiler allows storing only integer values into it. If you try to store a fractional value in an int type variable it will accept only the integer portion of the fractional value..

Illustration 9.1: C++ Program to get 2 integer numbers and display their sum:

```
// Program to receive two integer numbers and display their sum
#include <iostream>
using namespace std;
int main()
{
        int num1, num2, sum;
                        // variables num1, num2, and sum are declared as integers
        cout << "\n Enter Number 1: ";
        cin >> num1;
        cout << "\n Enter Number 2: ";
        cin >> num2;
        sum = num1 + num2;
        cout << "\n The sum of " << num1 << " and " << num2 << " is " << sum;
}
```

In the above program, there are three variables declared viz. **num1, num2** and **sum**. All these three variables are declared as integer types. So, three different integer values can be stored in these variables.

The variables **num1** and **num2** are used to store the values that are obtained from the user during the execution of the program, whereas the variable sum is used to store the processed (resultant) value.

During the execution of the above program, line numbers 7 and 9 prompt the user to Enter number 1 and number 2.